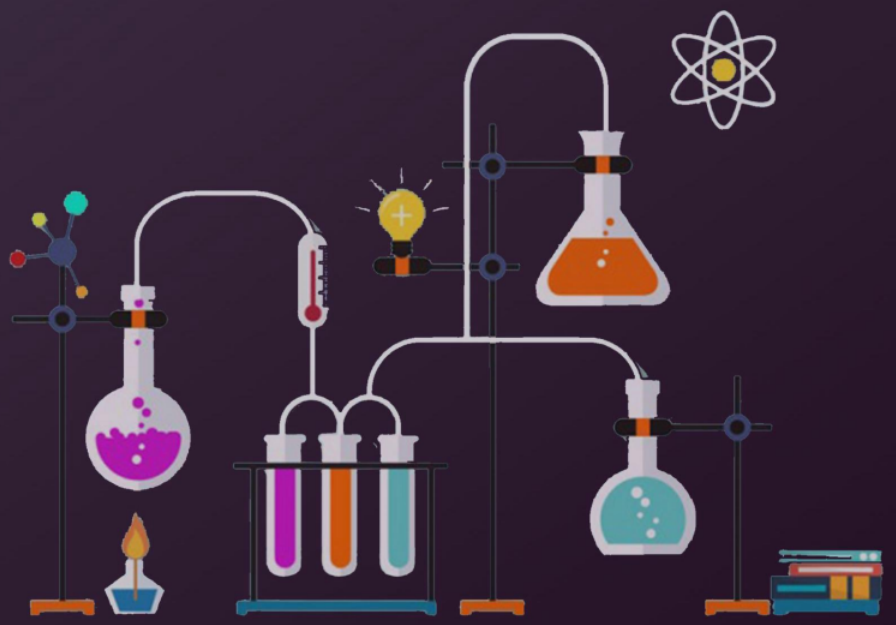
CHEMISTRY

Chemical Kinetics



SHEET

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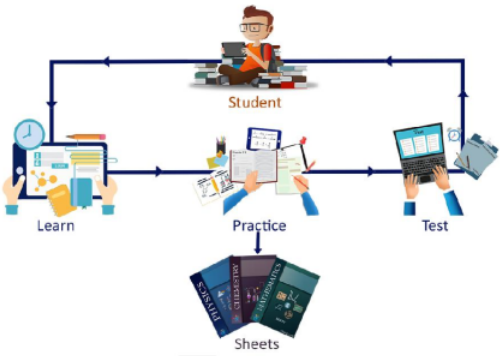
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Chemical Kinetics	1			
2	Rate of Reaction	2 - 8	7,8,24, 14-17	1-2	19
3	Rate Law, Order and Molecularity	9 - 16	1-3,5,9	3,8,10	1,2
4	Zero Order Reaction	17 - 23	12	4	
5	First Order Reaction	24 - 36	6,19-23	5-7,9	10-11, 15,20
6	Second Order Reaction	37 - 47	26,27		3,16
7	Methods of Determination of Order	48 - 49	4,11,18	11-15	6,14, 17,21
8	First Order Parallel, Sequential and Reversible Reactions	50		16	4,12-13
9	Activation Energy of a Reaction	51-57	13,28,29		5
10	Orientation Barrier and Arrhenius Equation	58-63	30-33	17-19	22,23
11	Factors Affecting Rate of Reaction	64-66			7,18
12	Rate Law From RDS and Steady State	67-70	10,25	20-22	4,8,9

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Indroduction to Chemical Kinetics

Q1. 'Kinetics' means :-

(A) Static (B) Moving

(C) Constant (D) None of these

Rate of Reaction

Q2. For the reaction, $2SO_2 + O_2 \longrightarrow 2SO_3$, the rate of disappearance of O_2 is 2.0×10^{-4} mol $L^{-1}s^{-1}$.

The rate of appearance of SO_3 is?

(A) 2.0×10^{-4} mol $L^{-1}s^{-1}$

(B) 4.0×10^{-4} mol $L^{-1}s^{-1}$

(C) 1.0×10^{-4} mol $L^{-1}s^{-1}$

(D) 6.0×10^{-4} mol $L^{-1}s^{-1}$

Q3. For a chemical reaction $3O_2 \rightarrow 2O_3$ the rate of formation of O_3 is 0.04 mole lit^{-1} sec^{-1}. Determine the rate of disappearance of O_2 :-

(A) 0.04 mol lit^{-1} sec^{-1}

(B) 0.08 mol lit^{-1} sec^{-1}

(C) 0.10 mol lit^{-1} sec^{-1}

(D) 0.06 mol lit^{-1} sec^{-1}

Q4. For the reaction : $N_2 + 3H_2 \rightarrow 2NH_3$

If the rate of disapperance of H_2 is $1.8 \times 10^{+3}$ mol $lit^{-1}sec^{-1}$. What is the rate of formation of ammonia-

(A) 0.9×10^3 (B) 1.2×10^3

(C) 2.7×10^3 (D) None

Q5. For a reaction $AB_5 \rightarrow AB + 4B$

The rate can be expressed in following terms

$$\frac{-d[AB_5]}{dt} = K\,[AB_5]\ ;\ \frac{d[B]}{dt} = K_1\,[AB_5].$$

So the correct relation between K and K_1 is :

(A) $K_1 = K$ (B) $K_1 = 2K$

(C) $K_1 = 4K$ (D) $2K_1 = K$

Q6. The rate of a reaction is expressed different in ways as follows :

$$-\frac{1}{3}\frac{d[D]}{dt} = \frac{-d[B]}{dt} = +\frac{1}{2}\frac{d[C]}{dt} = +\frac{1}{4}\frac{d[A]}{dt}$$

Therefore, the reaction is :-

(A) $4A + B \rightarrow 2C + 3D$

(B) $B + 3D \rightarrow 4A + 2C$

(C) $A + B \rightarrow C + D$

(D) $B + D \rightarrow A + C$

Q7. In a gaseous phase reaction, $A_2(g) \rightarrow B(g) + (1/2)\,C(g)$, the increase in pressure from 100 mm to 120 mm is noticed in 5 minute. The rate of disappearance of A_2 in mm min^{-1} is :

(A) 4 (B) 8

(C) 16 (D) 2

Q8. In a reaction $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$ the rate of appearance of NH_3 is 2.5×10^{-4} mol L^{-1} sec^{-1}. The rate of reaction & rate of disappearance of H_2 will be(In mol$L^{-1}sec^{-1}$)

(A) 3.75×10^{-4}, 1.25×10^{-4}

(B) 1.25×10^{-4}, 2.5×10^{-4}

(C) 1.25×10^{-4}, 3.75×10^{-4}

(D) 5.0×10^{-4}, 3.75×10^{-4}

Rate law, Order and Molecularity

Q9. The rate of certain hypothetical reaction $A + B + C \rightarrow$ products is given by

$$r = \frac{-d[A]}{dt} = K[A]^{1/2}[B]^{1/3}[C]^{1/4}$$ The order of the reaction –

(A) 1 (B) $\frac{1}{2}$

(C) 2 (D) $\frac{13}{12}$

Q10. For a reaction of the type A + B → products, it is observed that doubling the concentration of A causes the reaction rate to be four times, but doubling the amount of B does not effect the rate. The rate equation is –

(A) Rate = K [A][B]

(B) Rate = K $[A]^2$

(C) Rate = K$[A]^2[B]$

(D) Rate = K$[A]^2[B]^2$

Q11. For an elementary process

$2X + Y \rightarrow Z + W$, the molecularity is –

(A) 2 (B) 1

(C) 3` (D) Unpredictable

Q12. The specific rate constant of a first order reaction depends on the :–

(A) Conc. of the reactant

(B) Conc. of the product

(C) Time

(D) Temperature

Q13. An elementary reaction between A and B is a second order reaction. Which of the following rate equations must be correct ?

(A) $r = k[A]^2[B]^0$ (B) $r = k[A]^{3/2}[B]^{1/2}$

(C) $r = k[A]^0[B]^2$ (D) $r = k[A][B]$

Q14. For a hypothetical reaction,

$aA + bB \rightarrow$ Product, the rate law is

$r = K[A]^x[B]^y$ then :-

(A) (a + b) = (x + y)

(B) (a + b) < (x + y)

(C) (a + b) > (x + y)

(D) (a + b) may or may not be equal to (x + y)

Q15. For the reaction $CH_3COCH_3 + I_2 \xrightarrow{H^{\oplus}}$ Products, the rate is governed by the expression,

$r = K[CH_3COCH_3][H^+]$

The order w.r.t. iodine is :-?

(A) 1 (B) 2

(C) 0 (D) 3

Q16. For any reaction,

$A + B \longrightarrow C + D$

Rate law obtained is : Rate = k [A] [B]

Select the correct option(s) :

(A) Reaction must be elementary

(B) Reaction must be complex

(C) Reaction may be elementary or complex

(D) Reaction can not be complex

Zero Order Reaction

Q17. K for a zero order reaction is 2×10^{-2} $molL^{-1}$ sec^{-1}. If the concentration of the reactant after 25 sec is 0.5 M. Find the initial concentration of reactant.

(A) 0.5 M (B) 1.25 M

(C) 12.5 M (D) 1.0 M

Q18. For a zero order reaction, the plot of conc. (a – x) vs time is linear with :-

(A) +ve slope and zero intercept

(B) –ve slope and zero intercept

(C) +ve slope and non-zero intercept

(D) –ve slope and non-zero intercept

Q19. Consider a reaction A → B + C. If the initial concentration of A was reduced from 4M to 2M in 1hr and from 2M to 1M is 0.5 hour, the order of the reaction is :-?

(A) One (B) Zero

(C) Second (D) Third

Q20. If the initially concentration is reduced to 1/4th in a zero order reaction, the time taken for half of the reaction to complete :-

(A) Becomes one fourth

(B) Becomes 4 times

(C) Remains same

(D) Becomes double

Q21. If the rate of a gaseous reaction is independent of pressure the order is :?

(A) 0 (B) 1

(C) 2 (D) 3

Q22. For the reaction A → B, data of initial concentration and corresponding half life period are given in the tabular from :

[A]	1M	2M	4M
$T_{0.5}$	300s	600s	1200s

The order of the reaction is ?

(A) 0 (B) 1

(C) 2 (D) 3

Q23. In a certain reaction 10% of the reactant decomposes in one hour, 20% in two hours, 30% in three hours and so on. Dimension of the velocity constant (rate constant) are-

(A) Hour $^{-1}$

(B) Mol litre^{-1}hour $^{-1}$

(C) Litre mol^{-1} sec^{-1}

(D) Mol sec^{-1}

First Order Reaction

Q24. The reaction N_2O_5 (in CCl_4) → $2NO_2$ + ½ O_2 (g) is first order in N_2O_5 with rate constant $6.2 \times 10^{-4} S^{-1}$. What is the value of rate of reaction when $[N_2O_5]$ = 1.25 mole L^{-1} ?

(A) 7.75×10^{-4} mol L^{-1} S^{-1}

(B) 6.35×10^{-3} mol L^{-1} S^{-1}

(C) 5.15×10^{-5} mol L^{-1} S^{-1}

(D) 3.85×10^{-4} mol L^{-1} S^{-1}

Q25. In a first order reaction, the initial conc. of the reactant was M/10. After 8 minutes 20 seconds the conc. becomes M/100. What is the rate constant ?

(A) 5×10^{-3} second^{-1}

(B) 2.303×10^{-5} second^{-1}

(C) 2.303×10^{-4} second^{-1}

(D) 4.606×10^{-3} second^{-1}

Q26. In a first order reaction the concentration of the reactant is decreased from 1.0 M to 0.25 M in 20 min. The rate constant of the reaction would be ?

(A) 10 min^{-1} (B) 6.931 min^{-1}

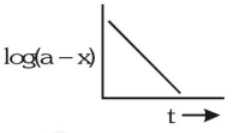
(C) 0.6931 min^{-1} (D) 0.06931 min^{-1}

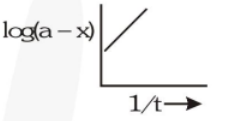
Q27. For a first order reaction A → products, the rate of reaction at [A] = 0.2 M is 1×10^{-2} mol lit^{-1}min^{-1}. The half life period for the reaction is ?

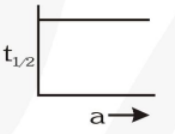
(A) 832 min. (B) 440 sec.

(C) 416 min. (D) 14 min.

Q28. Which of the following curves represents a Ist order reaction :–

(A) 

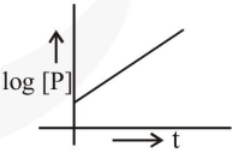
(B) 

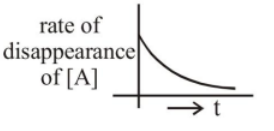
(C) 

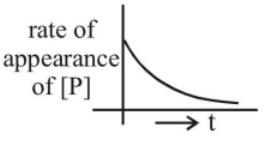
(D) 1 & 3 both

Q29. Which one of the following graph is incorrect depiction of first order process - "A → P"

(A) 

(B) 

(C) 

(D) 

Q30. For the first order reaction with rate constant K, which expression gives the half-life period (Initial concentration = a) :-

(A) $\frac{\ell n 2}{K}$ (B) $\frac{1}{Ka}$

(C) $\frac{0.693}{Ka}$ (D) $\frac{3}{2Ka^2}$

Q31. A first order reaction is 75% completed in 100 minutes. How long time will it take for it's 87.5% completion :-

(A) 125 min (B) 150 min

(C) 175 min (D) 200 min

Q32. What is the slope of graph plotted between half-life ($t_{1/2}$) and initial concentration (a) for a Ist order reaction:

(A) K (B) –1

(C) –K (D) Zero

Q33. The half life period of first order reaction is 69.3 min. Calculate the time taken 90% reaction to be complete :-

(A) 230 min (B) 230 sec

(C) 460 min (D) 460 sec

Q34. Wrong data for the first order reaction are ?

(A) $t_{50\%} = 100$ sec. , $t_{75\%} = 200$ sec

(B) $t_{75\%} = 32$ min , $t_{50\%} = 16$ min

(C) Both the above

(D) $t_{50\%} = 100$ sec. , $t_{75\%} = 150$ sec

Q35. The ratio of the time required for $\frac{3}{4}$ of the reaction of first order to complete to that required for half of the reaction is ?

(A) 1 : 1 (B) 1 : 2

(C) 2 : 1 (D) 3 : 2

Q36. Point out the wrong statement :

For a first order reaction

(A) Time for half-change ($t_{1/2}$) is independent of initial concentration

(B) Change in the concentration unit does not change the rate constant (K)

(C) Time for half-change × rate constant = 0.693

(D) The unit of K is mole^{-1} min^{-1}

Second Order Reaction

Q37. The acidic hydrolysis of ester is ?

(A) I order reaction

(B) Bimolecular reaction

(C) Pseudo unimolecular reaction

(D) All of these

Q38. For a given reaction, rate = $K[A]^1[B]^{2/3}$, the unit of rate constant K can be given as-

(A) mol$^{-2/3}$ litre$^{2/3}$ time

(B) mol$^{2/3}$litre$^{-2/3}$time^{-1}

(C) mol$^{-2/3}$ litre$^{2/3}$, time^{-1}

(D) mol$^{2/3}$ litre$^{-2/3}$, time

Q39. Rate of reaction A + B → Product, is given as a function of different initial concentration of A and B :-

(A) (mol L^{-1})	(B) (mol L^{-1})	Initial rate (mol L^{-1} min^{-1})
0.01	0.01	0.005
0.02	0.01	0.010
0.01	0.02	0.005

Determine the rate expression for the reaction -

(A) $r = k [A]^0 [B]^1$ (B) $r = k [A]^1 [B]^0$

(C) $r = k [A] [B]$ (D) $r = k [A]^2 [B]^0$

Q40. The given data are for the reaction :

$2NO(g) + Cl_2(g) \rightarrow 2NOCl(g)$ at 298 K

	$[Cl_2]$	[NO]	Initial rate (mol L^{-1} sec^{-1})
I	0.05 M	0.05M	1×10^{-3}
II	0.15 M	0.05M	3×10^{-3}
III	0.05 M	0.15M	9×10^{-3}

The rate law for the reaction is :

(A) rate = $k[NO][Cl_2]$

(B) rate = $k[Cl_2][NO]^2$

(C) rate = $k[Cl_2]^2[NO]$

(D) rate = $k[Cl_2]$

Q41. Which of the following rate law applies to a second order reaction involving only one reactant -

(A) Rate is equal to the square of concentration

(B) Rate is directly proportional to the square of concentration

(C) Rate is directly proportional to the square root of concentration

(D) None

Q42. For a second order reaction, if the conc. of a reactant decreases from 0.08 M to 0.04 M in ten minutes, what would be the time taken for the concentration to decrease to 0.01 M :-

(A) 20 min. (B) 30 min.

(C) 50 min. (D) 70 min.

Q43. If 'a' is the initial concentration of the reactant, the half-life period of the reaction of nth order is inversely proportional to -

(A) a^n (B) a^{n-1}

(C) a^{1-n} (D) a^{n+1}

Q44. Half life (t_1) of the first order reaction and half life (t_2) of the 2nd order reaction are equal. Hence ratio of the rate at the start of the reaction. (It initial concentration of reactants are 1M.)

(A) 1 (B) 2

(C) 0.693 (D) 1.44

Q45. For a hypothetical reaction,

$A + B \longrightarrow C + D$, the rate $= k[A]^{-1/2}[B]^{3/2}$. On doubling the concentration of A and B, the rate will be increased:

(A) 4 times (B) 2 times

(C) 3 times (D) 8 times

Q46. The rate of a gaseous reaction is given by the expression $r = k[A]^2[B]^3$. The volume of reaction vessel is suddenly reduced to one half of the initial volume. The reaction rate relative to the original rate will be ?

(A) 1/24 times (B) 1/32 times

(C) 32 times (D) 24 times

Q47. For a second order reaction $\frac{dx}{dt} = k(a-x)^2$.

Its half life period is ?

(A) $\frac{1}{a.k}$ (B) $\frac{0.693}{k}$

(C) $\frac{a}{k}$ (D) $\frac{0.693}{ak}$

Methods of Determination of Order

Q48. The rate constant of a reaction is 1.3×10^{-3} $Lit^2\ mol^{-2}\ min^{-1}$. The order of reaction is -

(A) zero (B) 1

(C) 2 (D) 3

Q49. Consider following two reaction,

$$A \longrightarrow \text{Product} ; \quad -\frac{d[A]}{dt} = k_1[A]^0$$

$$B \longrightarrow \text{Product} ; \quad -\frac{d[B]}{dt} = k_2[B]$$

Units of k_1 and k_2 are expressed in terms of molarity (mol L^{-1}) and time (sec^{-1}) as :-

(A) sec^{-1}, M sec^{-1}

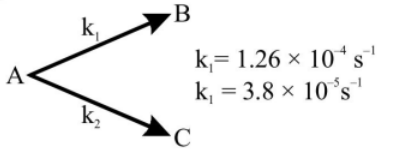
(B) M sec^{-1}, M sec^{-1}

(C) sec^{-1}, M^{-1} sec^{-1}

(D) M sec^{-1}, sec^{-1}

First Order Parallel, Sequential and Reversible Reactions

Q50. A Substance undergoes first order decomposition. The decomposition follows two parallel first order reactions as :



The percentage distribution of B and C are

(A) 80% B and 20% C

(B) 60% B and 40% C

(C) 90% B and 10% C

(D) 76.83% B and 23.17% C

Activation Energy of a Reaction

Q51. Activation energy of a reaction is ?

(A) The energy released during the reaction

(B) The energy evolved when activated complex is formed

(C) Minimum amount of external energy needed to overcome the potential barrier of reaction

(D) The energy needed to form one mole of the product

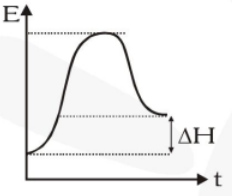
Q52. For the decomposition of N_2O_5, it is given that $2N_2O_5(g) \rightarrow 4NO_2(g)+O_2(g)$, activation energy(Ea)

$N_2O_5(g) \rightarrow 2NO_2(g) + 1/2\ O_2$, activation energy(Ea')

(A) Ea' > Ea (B) Ea' < Ea

(C) Ea' = Ea (D) Ea = 2E'a

Q53 For an endothermic reaction :



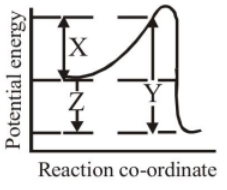
(A) $E_a > \Delta H$

(B) $E_a < \Delta H$

(C) $E_a = \Delta H$

(D) any one of the above

Q54. The potential energy diagram for reaction $A \rightarrow B$ is.



The activation energy of reaction is ?

(A) X (B) X + Z

(C) Y (D) X – Y

Q55. In order to react, a molecule at the time of collision must posses a certain amount of energy known as-

(A) Free energy

(B) Kinetic energy

(C) Threshold energy

(D) Internal energy

Q56. Which increases in increase of temperature

(A) Energy of activation

(B) Collision frequency

(C) Rate constant

(D) Both (B) and (C)

Q57. In respect of the equation $k = Ae^{-Ea/RT}$ in chemical kinetics, which one of the following statement is correct ?

(A) E_a is energy of activation

(B) R is Rydberg's constant

(C) k is equilibrium constant

(D) A is adsorption factor

Orientation Barrier and Arrhenius Equation Collision Theory

Q58. Arrhenius equation may be written as :–

(A) $\frac{d}{dT}(\ell n\ K) = -\frac{Ea}{RT}$

(B) $\frac{d}{dT}(\ell n\ K) = -\frac{Ea}{RT^2}$

(C) $\frac{d}{dT}(\ell n\ K) = +\frac{Ea}{RT^2}$

(D) $\frac{d}{dT}(\ell n\ K) = \frac{Ea}{RT}$

Q59. The plot of lnK versus 1/T is linear with a slope of :

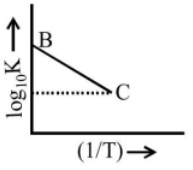
(A) $-E_a/R$

(B) E_a/R

(C) $-E_a/(2.303\ R)$

(D) $E_a/(2.303\ R)$

Q60. Following figure shows a graph in $\log_{10}K$ vs $= \frac{1}{T}$ where K is rate constant and T is temperature. The straight line BC has slope, $\tan\theta = -\frac{1}{2.303}$ and an intercept of 5 on Y-axis Thus E_a, the energy of activation is ?



(A) 2.303 × 2 cal (B) 2/2.303 cal
(C) 2 cal (D) None of these

Q61. The ΔH value of the reaction $H_2 + Cl_2 \rightarrow 2HCl$ is –44.12 kcal. If E_1 is the activation energy of the products, then for the above reaction :

(A) $E_1 > E_2$
(B) $E_1 < E_2$
(C) $E_1 = E_2$
(D) ΔH is not related to E_1 and E_2

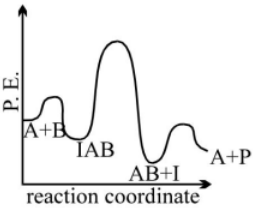
Q62. Collision theory is satisfactory for

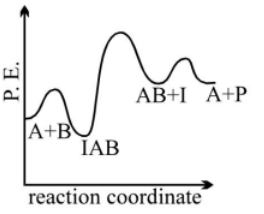
(A) First order reaction
(B) Second order reaction
(C) Bimolecular reactions
(D) Zero order reaction

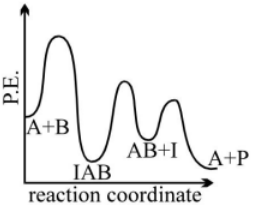
Q63. The following mechanism has been proposed for the exothermic catalyzed complex reaction.

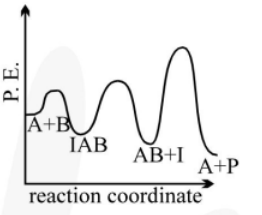
$$A + B \overset{\text{fast}}{\rightleftharpoons} I\,AB \xrightarrow{k_1} AB + I \xrightarrow{k_2} P + A$$

If k_1 is much smaller than k_2. The most suitable qualitative plot of potential energy (P.E.) versus reaction coordinate for the above reaction.

(A)


(B)


(C)


(D)


Factor Affecting Rate of Reaction

Q64. Catalyst in a chemical reaction :–

(A) Increase activation energy
(B) Does not change activation energy
(C) Does not change ΔH
(D) Decreases ΔH

Q65. A negative catalyst increases-

(A) Rate of reaction
(B) activation energy
(C) Ammount of product formed
(D) Rate of backward reaction

Q66. Which of the following explains the increase of the reaction rate by catalyst -

(A) Catalyst decreases the rate of backward reaction so that the rate of forward reaction increases
(B) Catalyst provides extra energy to reacting molecules so that they may reduce effective collisions
(C) Catalyst provides an alternative path of lower activation energy to the reactants
(D) Catalyst increases the number of collisions between the reacting molecules.

Rate law from Given Mechanism

Q67. Following mechanism has been proposed for a reaction.

$2A + B \rightarrow D + E$

$A + B \rightarrow C + D$ (slow)

$A + C \rightarrow E$ (fast)

The rate law expression for the reaction is –

(A) $r = K[A]^2[B]$ (B) $r = K[A][B]$

(C) $r = K[A]^2$ (D) $r = K[A][C]$

Q68. Steps of a multi-step reaction are as follows :-

(A) $A + B \xrightarrow{r_1 = 0.05} X$

(B) $X + B \xrightarrow{r_2 = 0.89} Y$

(C) $Y + A \xrightarrow{r_3 = 0.001} AY$

(D) $AY + B \xrightarrow{r_4 = 0.10} AYB$

Rate of reaction will be determined by :-

(A) Step (A) (B) Step (B)

(C) Step (C) (D) Step (D)

Q69. For the reaction $2H_2 + 2NO \rightarrow N_2 + 2H_2O$ the following mechanism has been proposed :-

(i) $2NO \rightleftharpoons N_2O_2$ (fast)

(ii) $N_2O_2 + H_2 \xrightarrow{k_2} N_2O + H_2O$ (slow)

(iii) $N_2O + H_2 \longrightarrow N_2 + H_2O$ (fast)

then what will be the rate law of this reaction ?

(A) $k[NO][H_2]^2$ (B) $k[NO][H_2]$

(C) $k[NO]^0[H_2]^3$ (D) $k[NO]^2[H_2]$

Q70. The chemical reaction

$2O_3 \longrightarrow 3O_2$, proceeds as follows :

$O_3 \rightleftharpoons O_2 + O$.....(fast)

$O + O_3 \longrightarrow 2O_2$.....(slow)

The overall order of the reaction is ?

(A) 1 (B) –1

(C) 0 (D) 2

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following are experimentally determined ?

(A) Rate constant (B) Rate law

(C) Order (D) Molecularity

Q2. Which of the following are not the permitted values of molecularity ?

(A) 3 (B) 0 (C) 2 (D) 4

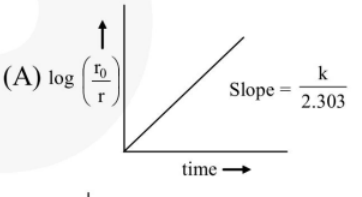
Q3. If the concentration of the reactants in the reaction $2A + B \rightarrow C + D$ is increased by three folds, the rate of the reaction will be increased by-

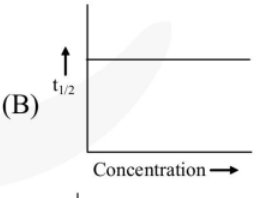
(A) 27 times (B) 9 times

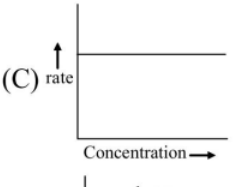
(C) 64 times (D) 01 times

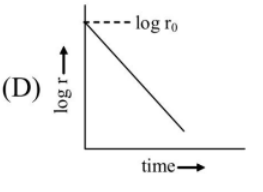
Q4. Which of the following plot represents first order reaction :

(r = rate at time 't' and r_0 is the initial rate).

(A) 

(B) 

(C) 

(D) 

Q5. Rate constant of a reaction depends on –
(A) Conc. of reactants (B) Temperature
(C) Nature of reactants (D) Time

Q6. Consider the following first order competing reactions :

$X \xrightarrow{k_1} A + B$ and $Y \xrightarrow{k_2} C + D$

If 50% of the reaction of X was completed when 96% of the reaction of Y was completed, the ratio of their rate constants (k_2/k_1) is :
(A) 4.64 (B) 0.215 (C) 1.1 (D) 4.65

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

(Paragraph For C(Que. 7 to 8)

For a hypothetical elementary reaction $A \xrightarrow{k_1} 2B$, $A \xrightarrow{k_2} 2C$

where $\frac{k_1}{k_2} = \frac{1}{2}$

Initially only 2 moles of A are present.

Q7. The total number of moles of A, B & C at the end of 50% reaction are ?
(A) 2 (B) 3
(C) 5 (D) None

Q8. Number of moles of B are ?
(A) 2 (B) 1
(C) 0.666 (D) 0.333

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q9. The decomposition of ammonia on platinum surface follow the change, $2NH_3 \rightarrow N_2 + 3H_2$;

$K = 2.5 \times 10^{-4}\ Ms^{-1}$ & $-\frac{d[NH_3]}{dt} = \frac{K_1[NH_3]}{1 + K_2[NH_3]}$, where K_1 and K_2 are constant.

Column-I	**Column-II**
(A) $\frac{-d[NH_3]}{dt}$	(P) Rate of formation of H_2
(B) $\frac{d[N_2]}{dt}$	(Q) 2.5×10^{-4} M sec^{-1}
(C) $\frac{d[H_2]}{dt}$	(R) Rate of decompositon of NH_3
(D) If the decomposition is zero order, the rate of production of N_2	(S) First
(E) If the decomposition is zero order, the rate of production of H_2	(T) Rate of formation of N_2
(F) The order for decomposition of NH_3 if $[NH_3]$ is very very less	(U) Zero
(G) The order for decomposition of NH_3 if $[NH_3]$ is very very high	(V) 7.5×10^{-4} M sec^{-1}

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q10. Column match :

Column-I	Column-II
(A) Elementary reaction	(p) High energy of activation
(B) Complex reaction	(q) Low energy of activation
(C) Slow reaction	(r) Several elementary steps
(D) Fast reaction	(s) Single step

(A) A→s; B→r ; C→p; D→q
(B) A→p; B→q; C→s; D→r
(C) A→s; B→p; C→q; D→r
(D) A→s; B→p; C→r; D→s

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q11. At 500 K, the half life period of a gaseous at an inital pressure of 80 kPa is 350 sec. When the pressure is 40 kPa, the half life period is 175 sec, the order of the reaction is ?

Q12. A drop of solution of volume 0.05 ml contains 3×10^{-6} mole of H^+. If the rate constant for the disappearance of H^+ is 10^7 mol lit^{-1} s^{-1}. How many nano seconds it required for H^+ in drop to disappear ?

Q13. In gaseous reactions important for understanding the upper atmosphere, H_2O and O react biomolecularly to form two OH radicals. ΔH for this reaction is 72 kJ at 500 K and E_a =77 kJ mol^{-1}, then calculate E_a for the biomolecular recombination of 2OH radicals to form H_2O & O at 500 K.

Q14. For the reaction : $2N_2O_5 \rightarrow 4NO_2 + O_2$. If the concentration of NO_2 increases by 1.6×10^{-2} M in 4s. Calculate the following -
(i) rate of formation of NO_2
(ii) rate of formation of O_2
(iii) rate of disappearance of N_2O_5
(iv) rate of reaction

Q15. Express rate of reaction for the following changes :
(a) $2HI \rightarrow H_2 + I_2$
(b) $2SO_2 + O_2 \rightarrow 2SO_3$

Q16. For the decomposition reaction :
$N_2O_{4(g)} \rightarrow 2NO_{2(g)}$; the initial pressure of N_2O_4 falls from 0.46 atm to 0.28 atm in 30 minute. What is the rate of appearance of NO_2 ?

Q17. The following was studied in a closed vessel
$2N_2O_5(g) \rightleftharpoons 4NO_2(g) + O_2(g)$
It was found that the concentration of NO_2 increases by 2.0×10^{-2} mol L^{-1} in five seconds.
Calculate :

(i) the rate of reaction

(ii) the rate of change of concentration of N_2O_5

Q18. For a reaction ; 3A $\rightarrow$ Products, it is found that the rate of reaction doubles if concentration of A is increased four times, calculate order of reaction.

Q19. 90% of a first order reaction was completed in 10 hours. When will 99.9% of the reaction complete?

Q20. Show that time required to complete 99.9% completion of a first order reaction is about three times to 90% completion.

Q21. 20% of a first order reaction was completed in 5 min. When will 60% of the reaction complete?

Q22. For the first order reaction, half life is 6 min. Calculate the rate constant of the reaction ?

Q23. If the decomposition of nitrogen (V) oxide $2N_2O_5 \longrightarrow 4NO_2 + O_2$, following a first order kinetics,

(i) Calculate the rate constant for a 0.04 M solution, if the instantaneous rate is 1.4×10^{-6} mol L^{-1} s^{-1}.

(ii) Also calculate the rate of reaction when the concentration of N_2O_5 is 1.20 M.

(iii) What concentration of N_2O_5 would give a rate of 2.45×10^{-5} mol L^{-1} s^{-1} ?

Q24. In the formation of sulphur trioxide by the contact process $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$

The rate of reaction is expressed as $-\frac{d(O_2)}{dt}$ $= 2.5 \times 10^{-4}$ mol L^{-1} sec^{-1}. The rate of disappearance of (SO_2) will be –

Q25. Give the following data for the reaction : A + B $\rightarrow$ Product.

Experiment No.	[A]	[B]	rate
1	1	2	4
2	2	2	4
3	2	4	16

What is the rate law equation ?

Q26. For a second order reaction in which both the reactants have equal initial concentration, the time taken for 60% completion of reaction is 3000 second. What will be the time taken for 20% of the reaction ?

Q27. The reaction $2A + B + C \rightarrow D + E$ is found to be first order in A second in B and zero order in C.

(i) Give the rate law for the reaction in the form of differential equation.

(ii) What is the effect in rate of increasing concentrations of A, B and C two times?

Q28. The E_a of exothermic reaction A $\rightarrow$ B is 80 kJ/mol. Heat of reaction is 20 kJ/mol. E_a for the reaction B$\rightarrow$A will be ?

Q29. For the reaction A + B $\rightleftharpoons$ C + D the activation energy is 32 kJ/mol. For the reverse reaction the E_a is 58 kJ/mol. Determine (i) Nature of reaction (ii) ΔH

Q30. For a reaction, temperature coefficient = 2, then calculate the activation energy (in KJ) of the reaction.

Q31. For first order gaseous reaction log k when plotted against $\frac{1}{T}$ it give a straight line with aslope of –8000. Calculate the activation energy of the reaction.

Q32. If temperature of a reaction is increased from 10°C to 100°C then how many times rate of reaction will become?

Q33. A reaction is carried out at 10°C. If temperature is increased by 50°C then how many times rate of reaction will become?

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. In acidic medium the rate of reaction between $(BrO_3)^-$ and Br^- ion is given by the expression $\frac{d[BrO_3^-]}{dt} = k[BrO_3^-][Br^-][H^+]^2$

it means :

(A) Rate constant of overall reaction is 4 sec^{-1}.

(B) Rate of reaction is independent of the conc. of acid.

(C) The change in pH of the solution will not affect the rate.

(D) Doubling the conc. of H^+ ions will increase the reaction rate by 4 times.

Q2. The reduction of peroxydisulphate ion by I^- ion is expressed by

$S_2O_8^{2-} + 3\,I^- \longrightarrow 2SO_4^{2-} + I_3^-$.

If rate of disappearance of I^- is $9/2 \times 10^{-3}$ mol lit^{-1} S^{-1}, what is the rate of formation of SO_4^{2-} during same time?

(A) 10^{-3} mol/L^{-1} S^{-1}

(B) 2×10^{-3} mol L^{-1} S^{-1}

(C) 3×10^{-3} mol L^{-1} S^{-1}

(D) 4×10^{-3} mol L^{-1} S^{-1}

Q3. Which of the following statement is wrong about order of reaction ?

(A) Order is experimentally determined

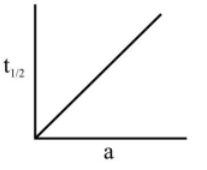
(B) Order is independent on the stoichiometry of reaction

(C) Order may not be fractional

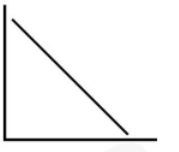
(D) Order is the sum of the power to which the concentration terms are raised to express the rate of reaction

Q4. Consider the reaction A $\longrightarrow$ B, graph between half life $(t_{1/2})$ and initial concentration (a) of the reactant is ?

Hence graph between $-\frac{d[A]}{dt}$ and time will be :



(A) $-\frac{d[A]}{dt}$



(B) $-\frac{d[A]}{dt}$



(C) $-\frac{d[A]}{dt}$



(D) $-\frac{d[A]}{dt}$



Q5. The reaction $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$ is first order with respect to N_2O_5. Which of the following graph would yield a straight line :

(A) $\log(P_{N_2O_5})$ versus time with – Ve slope

(B) $(P_{N_2O_5})^{-1}$ versus time

(C) $(P_{N_2O_5})$ versus time

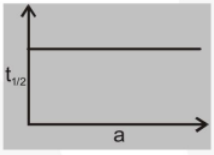
(D) $\log(P_{N_2O_5})$ versus time with + Ve slope

Q6. The expression which gives $\frac{1}{4}$ th life of Ist order reaction is :–

(A) $\frac{K}{2.303}\log\frac{4}{3}$ (B) $\frac{2.303}{K}\log 3$

(C) $\frac{2.303}{K}\log\frac{3}{4}$ (D) $\frac{2.303}{K}\log\frac{4}{3}$

Q7. A graph between $t_{1/2}$ and conc. for nth order reaction is a straight line. Reaction of this nature is completed 50% in 10 minutes when conc. is 2 mol L^{-1}. This is decomposed 50% in t minutes at 4 mol L^{-1} than n and t are respectively.



(A) 0, 20 min.

(B) 1, 10 min.

(C) 1, 20 min.

(D) 0, 5 min.

Q8. Consider the reaction, 2A + B → Product. When concentration of A alone was doubled, the half life did not change. When the concentration of B alone was doubled, the rate increased by two times. The unit of rate constant for this reaction is -

(A) L mol^{-1} s^{-1} (B) no unit

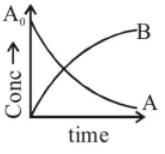
(C) mol L^{-1} s^{-1} (D) s^{-1}

Q9. The gaseous reaction A (g) ⟶ 2B (g) + C (g) is found to be first order with respect to A. If the reaction is started with $p_A = 90$ torr the pressure after 10 min is found to be 180 torr. The rate constant of the reaction is -

(A) 1.15×10^{-3} s^{-1} (B) 2.30×10^{-3} s^{-1}

(C) 3.45×10^{-3} s^{-1} (D) 4.60×10^{-3} s^{-1}

Q10. For the reaction, A ⟶ nB. At the point of intersection of the two curves, the concentration of B is given by .....



(A) $\frac{nA_0}{2}$ (B) $\frac{A_0}{2-1}$

(C) $\frac{nA_0}{n+1}$ (D) $\left(\frac{n-1}{n+1}\right)A_0$

Q11. Consider the plots, given below, for the type of reaction :-

$A \rightarrow B + C$



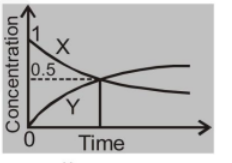
(I) (II) (III)

The plots respectively corresponds to the reaction order :-

(A) 0,1,2 (B) 1,2,0

(C) 1,0,2 (D) None of these

Q12. The accompanying figure depicts the change in concentration of species X and Y for the reaction X → Y as a function of time the point of intersection of the two curves reperesents.



(A) $t_{1/2}$

(B) $t_{3/4}$

(C) $t_{2/3}$

(D) Data insufficient to predict

Q13. Given that for a reaction of order n, the integrated form of the rate equation is

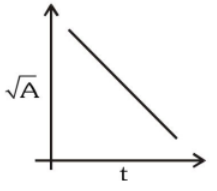
$$k = \frac{1}{t(n-1)}\left[\frac{1}{C^{n-1}} - \frac{1}{C_0^{n-1}}\right]$$

Where C_0 and C are the values of the reactant concentration at the start and after time t. What is the relationship between $t_{3/4}$ and $t_{1/2}$ where $t_{3/4}$ is the time required for C to become 1/4 C_0?

(A) $t_{3/4} = t_{1/2}[2^{n-1} + 1]$ (B) $t_{3/4} = t_{1/2}[2^{n-1} - 1]$

(C) $t_{3/4} = t_{1/2}[2^{n+1} - 1]$ (D) $t_{3/4} = t_{1/2}[2^{n+1} + 1]$

Q14. For the reaction A → B, the rate law expression is $-\frac{d[A]}{dt} = k[A]^{1/2}$. If initial concentration of [A] is $[A]_0$, then :



(A) the integerated rate expression is $k = \frac{2}{t}(A_0^{1/2} - A^{1/2})$

(B) The graph of $\sqrt{A}$ Vs t will be

(C) the half life period $t_{1/2} = \frac{K}{2[A]_0^{1/2}}$

(D) The time taken for 75% completion of reaction $t_{3/4} = \frac{\sqrt{[A]_0}}{k}$

Q15. The rate constant of n_{th} order has units -

(A) $Litre^{1-n}\ mol^{1-n}\ sec^{-1}$

(B) $Mol^{1-n}\ litre^{1-n}\ sec$

(C) $Mol^{1-n^2}\ litre^{n^2}\ sec^{-1}$

(D) $Mole^{1-n}\ litre^{1-n}\ sec^{-1}$

Q16. In the following first order competing reactions;

$A + Reagent \xrightarrow{k_1} product$

$B + Reagent \xrightarrow{k_2} Product$

The ratio of k_1 / k_2 if only 50% of B will have been reacted when 94% of A has been reacted is-

(A) 3 (B) 2 (C) 1 (D) 4

Q17. The incorrect statement is-

(A) All the collisions between reactant molecules do not lead to a chemical change.

(B) A zero order reaction proceeds at a constant rate independent of concentration or time.

(C) Fast reactions have low activation energies.

(D) In a first order reaction, the reaction ideally takes finite time to be completed.

Q18. The rate constant of a reaction is 1.5×10^{-3} at 25°C and 2.1×10^{-2} at 60°C. The activation energy is-

(A) $\frac{35}{333} R \log_e \frac{2.1\times10^{-2}}{1.5\times10^{-2}}$

(B) $\frac{298\times333}{35} R \log_e \frac{21}{1.5}$

(C) $\frac{298\times333}{35} R \log_e 2.1$

(D) $\frac{298\times333}{35} R \log_e \frac{2.1}{1.5}$

Q19. For the equilibrium, A(g) ⇌ B(g), ΔH is – 40 kJ/mol. If the ratio of the activation energies of the forward (E_f) and reverse (E_b) reactions is $\frac{2}{3}$ then :-

(A) E_f = 60 kJ/mol; E_b = 100 kJ/mol

(B) E_f = 30 kJ/mol; E_b = 70 kJ/mol

(C) E_f = 80 kJ/mol; E_b = 120 kJ/mol

(D) E_f = 70kJ/mol; E_b = 30 kJ/mol

Q20. For a chemical reaction Y + 2Z → Product, rate controlling step is?

$$Y + \frac{1}{2}Z \longrightarrow Q$$

If the concentration of Z is doubled, the rate of reaction will

(A) remain the same

(B) become four times

(C) become 1.414 times

(D) become double

Q21. Consider this gas phase reaction

$Cl_2(g) + CHCl_3(g) \longrightarrow HCl(g) + CCl_4(g)$

The reaction is found experimentally to follow this rate law.

$rate = k[CHCl_3][Cl_2]^{1/2}$

Based on this information, what conclusions can be drawn about this proposed mechanism ?

Step 1. $Cl_2(g) \longrightarrow 2Cl(g)$

Step 2. $Cl(g) + CHCl_3(g) \longrightarrow HCl(g) + CCl_3(g)$

Step 3. $Cl(g) + CCl_3(g) \longrightarrow CCl_4(g)$

(A) Step 1 is the rate-determining step

(B) Step 2 is rate-determining step

(C) Step 3 is the rate-determining step

(D) The rate-determining step cannot be identified

Q22. Consider this reaction :

$2\,NO_2\,(g) + O_3\,(g) \longrightarrow N_2O_5\,(g) + O_2\,(g)$

The reaction of NO_2 (g) and O_3 (g) represented is first order in NO_2 (g) and in O_3 (g) Which of the following mechanism is/are consistent with the rate law ?

Mechanism I :

$NO_2\,(g) + O_3\,(g) \longrightarrow NO_3\,(g) + O_2\,(g)$ (slow)

$NO_3\,(g) + NO_2\,(g) \longrightarrow N_2O_5\,(g)$ (fast)

Mechanism II :

$O_3\,(g) \rightleftharpoons O_2\,(g) + O$ (fast)

$NO_2\,(g) + O \longrightarrow NO_3\,(g)$ (slow)

$NO_3\,(g) + NO_2\,(g) \longrightarrow N_2O_5\,(g)$ (fast)

(A) I only (B) II only

(C) Both I & II (D) Neither I nor II

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which statements about the order of reaction is/are incorrect ?

(A) The order of reaction must be a positive integer

(B) A second order reaction is also bimolecular

(C) The order of reaction increases with increasing temperature.

(D) The order of reaction can only be determined by experiment

Q2. For which of the following reactions, unit of rate constants involves concentration as well as time.

(A) Zero order

(B) First order

(C) Fraction order reaction

(D) Negative order reaction

Q3. $Zn + 2H^+ \longrightarrow Zn^{2+} + H_2$; Half-life period is independent of concentration of zinc at constant pH. For the constant concentration of Zn, rate becomes 100 times when pH is decreased from 3 to 2. Hence :

(A) $\frac{dx}{dt} = k\,[Zn]^0[H^+]^2$

(B) $\left(\frac{dx}{dt}\right) = k\,[Zn][H^+]^2$

(C) Rate is not affected if concentration of zinc is made four times and that of H+ ion is halved.

(D) Rate becomes four times if concentration of H^+ ion is doubled at constant Zn concentration.

4. Which of the following reactions will have fractional order for A_2 or B_2 ?

(A) A_2 ⇌ $A + A$ (fast)

$A + A_2$ → $AB + B$ (slow)

$A + B$ ⇌ AB (fast)

(B) A_2 ⇌ C (slow)

$C + B_2$ ⇌ D (fast)

$D + A_2$ ⇌ Products

(C) B_2 ⇌ $B + B$ (fast)

$A_2 + B$ ⇌ $AB + A$ (slow)

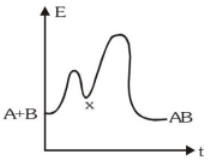
AB ⇌ Products (fast)

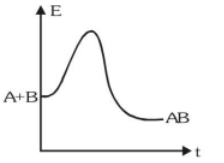
(D) All have fractional order

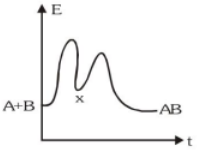
Q5. For an exothermic chemical process is occuring in two steps as follows

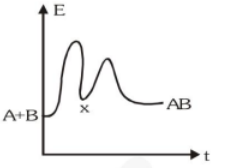
(i) $A + B \rightarrow X$ (slow) (ii) $X \rightarrow AB$ (fast)

the process of reaction can be best described by -

(A) 

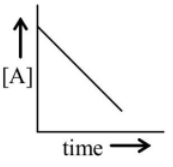
(B) 

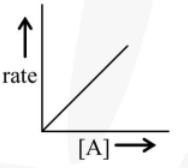
(C) 

(D) 

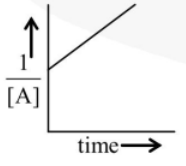
Q6. Which of the following is/are correctly matched?

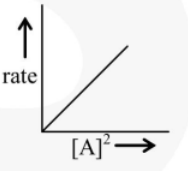
(A) For zero order reaction: $A \rightarrow$ Products (B) For first order reaction: $A \rightarrow$ Products





(C) For second order reaction: $A \rightarrow$ products (D) For second order reaction: $A \rightarrow$ products





Q7. Which of the following are incorrect ?

(A) total collision rate $\propto$ mean speed $\propto$ absolute temperature

(B) total collision rate $\propto \dfrac{1}{\text{mean speed}} \propto \dfrac{1}{\text{absolute temperature}}$

(C) total collision rate $\propto$ mean speed $\propto$ (absolute temperature)$^{1/2}$

(D) total collision rate $\propto$ (mean speed)^2 $\propto$ (absolute temperature)^3

Q8. Which of the following is/are **correct** statement?

(A) Stoichiometry of a reaction tells about the order of the elementary reactions.

(B) For a zero order reaction, rate and the rate constant are identical.

(C) A zero order reaction is controlled by factors other than concentration of reactants.

(D) A zero order reaction is always elementary reaction.

Q9. For the hypothetical reaction, $A_2 + B_2 \rightarrow 2AB$, the mechanism is given as below :

$A_2 \rightleftharpoons A + A$ (fast reaction)

$A + B_2 \longrightarrow AB + B$ (slow reaction)

$A + B \longrightarrow AB$ (fast reaction)

Then :

(A) The rate determining step is $A + B_2 \rightarrow AB + B$

(B) The order of the reaction is 3/2

(C) The overall molecularity is 4

(D) The rate expression is Rate = k [A] [B_2]

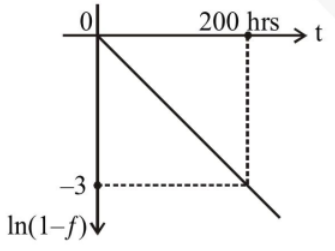
PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 10 to 11

Oxidation of metals is generally a slow electrochemical reaction involving many steps. These steps involve electron transfer reactions. A particular type of oxidation involve overall first order kinetics with respect to fraction of unoxidised metal **(1 – *f*)** surface thickness relative to maximum thickness **(T)** of oxidixed surface, when metal surface is exposed to air for considerable period of time :

Rate law : $\frac{df}{dt} = k(1 - f)$, **where** $f = x/T$,



x = thickness of oxide film at time 't' & T = thichness of oxide film at t = ∞

A graph of ln(1 – *f*) vs t is shown in the adjacent figure.

Q10. The time taken for thickness to grow 50% of 'T' is

(A) 23.1 hrs (B) 46.2 hrs

(C) 100 hrs (D) 92.4 hrs.

Q11. The exponential variation of '*f*' with t(hrs) is given by

(A) $[1 - e^{-3t/200}]$ (B) $e^{-3t/200} - 1$

(C) $e^{-3t/200}$ (D) $e^{3t/200}$

Paragraph for Question Nos. 12 & 13

For the given sequential reaction

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

the concentration of A, B & C at any time 't' is given by

$$[A]_t = [A]_0 e^{-k_1 t} \ ;$$

$$[B]_t = \frac{k_1[A]_0}{(k_2 - k_1)} \left[e^{-k_1 t} - e^{-k_2 t}\right]$$

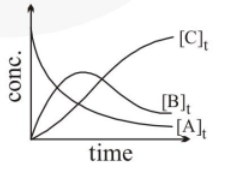
$$[C]_t = [A_0] - ([A]_t + [B]_t)$$

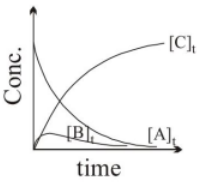
Q12. The time at which concentration of B is maximum is

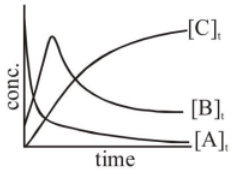
(A) $\frac{k_1}{k_2 - k_1}$ (B) $\frac{1}{k_2 - k_1} \ln \frac{k_1}{k_2}$

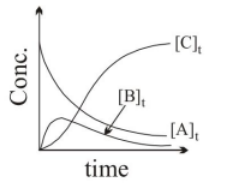
(C) $\frac{1}{k_1 - k_2} \ln \frac{k_1}{k_2}$ (D) $\frac{k_2}{k_2 - k_1}$

Q13. Select the correct option if $k_1 = 1000\ s^{-1}$ and $k_2 = 20\ s^{-1}$.

(A) 

(B) 

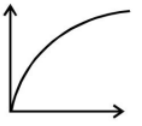
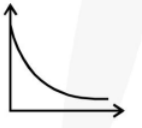
(C) 

(D) 

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q14. Match the column for the reaction of type $A(g) \longrightarrow 2B(g)$

List I	**List II**
(A) [A] vs t for zero order	(p) 
(B) $\frac{-d[A]}{dt}$ vs t for first order	(q) 
(C) [B] vs t for first order	(r) 
(D) $\frac{d[B]}{dt}$ vs $\frac{-d[A]}{dt}$ for zero order	(s) 

(A) a–r, b–s, c-q, d–p
(B) a–q, b–r, c–p, d–s
(C) a–s, b–r, c–q, d–p
(D) a–q, b–s, c–r, d–p

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q15. Match the column

Column I	**Column II**
(A) $t_{3/4} = \frac{3}{2} \times t_{1/2}$	(P) 99% completion
(B) $t_{3/4} = 2 \times t_{1/2}$	(Q) zero order
(C) $t = 4.606/k$	(R) first order
(D) $t = 2.303/k$	(S) 90% completion

Q16.

Column-A	**Column-B**
(A) First order	(P) Unit of k = sec^{-1}
(B) Pseudo first order	(Q) Unit of k = $dm^6\ mol^{-2}\ s^{-1}$
(C) Third order	(R) $t_{1/2} \propto 1/a^2$
(D) Second order	(S) $t_{1/2} \propto 1/a$

Q17. Match the Column :

Column-I **(Type of PLOT for reaction A → products)**	**Column-II**
(A) Rate (y-axis) Vs $[A]_t$ (x-axis) for second order reaction	(P) Non-linear
(B) Half-life (y-axis) Vs initial concentration of A(x-axis) for second order reaction	(Q) Positive slope
(C) Rate (y-axis) Vs $\frac{1}{[A]_t}$ (x-axis) for zero order reaction	(R) Zero slope
(D) log (Rate) (y-axis) Vs time (x-axis) for first order reaction	(S) Linear

Q18. Match the column

Column I	**Column II**
(A) Degree of dissociation $\left(\frac{x}{a}\right)$ for a first order reaction	(P) $r = PZe^{-E_a/RT}$
(B) Fraction of molecules having energy equal to or greater than activation energy	(Q) $\frac{d \ln k}{dt} = \frac{E_a}{RT^2}$
(C) Arrhenius equation	(R) $1 - e^{-kt}$
(D) Relationship between rate of reaction and collision frequency	(S) $\ln k = \ln A - \frac{E_a}{RT}$
	(T) $e^{-E_a/RT}$
	(u) $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$

SUBJECTIVE TYPE

Q19. For the decomposition, $N_2O_{5(g)} \rightarrow N_2O_{4(g)} + 1/2O_{2(g)}$, the initial pressure on N_2O_5 is 114 mm and after 20 s, the pressure of reaction mixture becomes 133 mm. Calculate the rate of reaction in terms of (a) change in atm s^{-1} and(b) change in molarity s^{-1}. Given that reaction is carried out at 127°C.

Q20. At 373 K, a gaseous reaction A ⟶ 2B + C is observed to be of first order. On starting with pure A it was found that at the end of 10 minutes the total pressure of the system was 176 mm of mercury and after a long time, when dissociation of A was complete, it was 270 mm. From these data, calculate :

(i) the initial pressure of A

(ii) the pressure of A at the end of 10 minutes.

(iii) the rate constant.

Q21. The decomposition of dimethyl ether leads to the formation of CH_4, H_2 and CO and the reaction rate is given by :

$$\text{Rate} = k[CH_3OCH_3]^{3/2}$$

The rate of reaction is followed by increase in pressure in a closed vessel and the rate can also be expressed in terms of the partial pressure of dimethyl ether i.e.,

$$\text{Rate} = k[P_{CH_3OCH_3}]^{3/2}$$

If the pressure is measured in bar and time in minutes, then what are the units of rate and rate constants?

Q22. For a reaction, temperature coefficient = 2, then calculate the activation energy (in KJ) at $T_1 = 25°C$ and $T_2 = 35°C$ of the reaction.

Q23. For the reaction $CO(g) + Cl_2(g) \rightarrow COCl_2(g)$ under the same concentration conditions of the reactants, the rate of the reaction at 250°C is 1500 times as fast as the same reaction at 150°C. Calculate the activation energy of the reaction. If the frequency factor is 2.0×10^{10} M^{-1} sec^{-1}, calculate the rate constant of the reaction at 150°C.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
B	B	D	B	C	B	B	C	D	B
11	12	13	14	15	16	17	18	19	20
C	D	D	D	C	C	D	D	B	A
21	22	23	24	25	26	27	28	29	30
A	A	B	A	D	D	D	D	B	A
31	32	33	34	35	36	37	38	39	40
B	D	A	D	C	D	D	C	B	B
41	42	43	44	45	46	47	48	49	50
B	D	B	C	B	C	A	D	D	D
51	52	53	54	55	56	57	58	59	60
C	C	A	A	C	D	A	C	A	C
61	62	63	64	65	66	67	68	69	70
A	C	A	C	B	C	B	C	D	A

Ex-1A

1. A, B, C **2.** B, D **3.** A **4.** A, B, D **5.** B, C

6. A **7.** A **8.** C

9. (A) → R ; (B) → T ; (C) → P ; (D) → Q ; (E) → V ; (F) → S ; (G) → U

10. A **11.** 0 **12.** 6 **13.** 5

14. (i) 4.0×10^{-3} M/s (ii) 1.0×10^{-3} M/s (iii) 2.0×10^{-3} M/s (iv) 1.0×10^{-3} M/s

15. (i) $\frac{d[I_2]}{dt}$ (ii) $-\frac{1}{2}\frac{d(SO_3)}{dt}$ **16.** 1.2×10^{-2} atm min^{-1}

17. (i) 10^{-3} (ii) 2×10^{-3} **18.** 1/2 **19.** 30 hour **21.** 20.54 min

22. 0.1155 min^{-1}

23. (i) 3.5×10^{-5} s^1 (ii) 4.2×10^{-5} mol L^{-1} s^{-1} (iii) 0.7 M

24. 5×10^{-4} mol L^{-1} s^{-1} **25.** 2 **26.** 500 sec

27. (i) $r = k[A]\,[B]^2\,[C]^0$ (ii) 8 **28.** 100 kJ/ mole **29.** -26 kJ/ mole (Exothermic)

30. 52.31 kJ **31.** 36608 calories **32.** 2^9 times

33. 32 times

Ex-2

1	2	3	4	5	6	7	8	9	10
D	C	C	C	A	D	B	A	A	C
11	12	13	14	15	16	17	18	19	20
C	A	A	A	D	D	D	B	C	C
21	22								
B	A								

Ex-2A

1. A, B, C **2.** A, C, D **3.** A, D **4.** A, C **5.** C

6. A, B, C, D **7.** A, B, D **8.** A, B, C **9.** A, B **10.** B

11. A **12.** C **13.** C **14.** B

15. A→ Q ; B → R ; C → P,R ; D → R,S

16. A → P ; B → Q ; C → Q, R ; D → S

17. (A)→(P); (B)→(P); (C)→(S,R); (D)→(S)

18. A→ r ; B→ t ; C→ q,s,u ; D→ p

19. 7.61×10^{-5} M ; 2.5×10^{-3} atm s^{-}

20. (i) 90 mm (ii) 47 mm (iii) 6.496 10^{-2} min^{-1}

21. $bar^{-1/2}$ min^{-1} **22.** 52.31 kJ **23.** 4.69×10^{-7}

-: Important Notes & Formulas :-